Research Papers

1. The Faiss Library

- **Authors:** Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, Hervé Jégou
- **Year:** 2024
- **Venue:** arXiv
- **Abstract:** Presents FAISS, a leading toolkit for vector similarity search supporting a spectrum of indexing methods (flat search, IVF, PQ, HNSW, etc.), compression, clustering, and GPU acceleration. The paper outlines design principles, optimization strategies, and benchmarks demonstrating efficiency and versatility across ML and retrieval tasks arxiv.org+2reddit.com+2reddit.com+2.
- **URL:** https://arxiv.org/abs/2401.08281

2. Optimizing Retrieval-Augmented Generation: Analysis of Hyperparameter Impact on Performance and Efficiency *(Technique/System Focus)*

- **Authors:** Adel Ammar, Anis Koubaa, Omer Nacar, Wadii Boulila
- **Year:** 2025
- **Venue:** arXiv
- **Abstract:** An empirical study of RAG, analyzing hyperparameters across vector stores (Chroma vs. FAISS), chunking strategies, re-ranking, and temperature. Measures seven metrics including retrieval precision/recall and generation performance. Finds that Chroma is ~13% faster while FAISS offers higher precision, and re-ranking improves quality but at a runtime cost. Offers clear recommendations for balancing speed and accuracy in real deployments .
- **URL:** https://arxiv.org/abs/2505.08445

3. RAGCache: Efficient Knowledge Caching for Retrieval-Augmented Generation *(System Contribution)*

- **Authors:** Chao Jin, Zili Zhang, Xuanlin Jiang, Fangyue Liu, Xin Liu, Xuanzhe Liu, Xin Jin

- **Year:** 2024
- **Venue:** arXiv
- **Abstract:** Proposes RAGCache, a multilevel caching system optimizing RAG inference by caching intermediate knowledge tree states across GPU and host. Introduces LLM-aware cache replacement and overlaps retrieval & generation. Integrated with vLLM and FAISS, the system achieves up to 4× faster time-to-first-token and 2.1× higher throughput .
- **URL:** https://arxiv.org/abs/2404.12457

4. Survey of Vector Database Management Systems *(Comparative Review)*

- **Authors:** James Jie Pan, Jianguo Wang, Guoliang Li
- **Year:** 2024
- **Venue:** *VLDB Journal* (Q1 journal)
- **Abstract:** A comprehensive evaluation of over 20 vector DBMSs (Chroma, FAISS, pgvector, Weaviate, Milvus, etc.). It identifies five key vector data challenges and analyzes state-of-the-art techniques in query processing, indexing (quantization, navigable graph structures), hybrid query operators, hardware acceleration, and benchmark practices. It also outlines open challenges and future directions arxiv.org+11colab.ws+11arxiv.org+11researchgate.net+1ui.adsabs.harvard.edu+1.
- **URL:** https://doi.org/10.1007/s00778-024-00864-X

5. When Large Language Models Meet Vector Databases: A Survey

- **Authors:** Zhi Jing, Yongye Su, Yikun Han
- **Year:** 2024
- **Venue:** arXiv
- **Abstract:** Reviews the synergy between LLMs and vector DBs, pinpointing the role of VecDBs in addressing hallucinations, outdated knowledge, cost, and memory limits in LLMs. Summarizes encoder architectures, vector indexing, retrieval methods, and system-level integration for RAG, and discusses emerging trends and integration challenges ivysci.com+3researchgate.net+3springerprofessional.de+3reddit.com+5arxiv.org+5cs.purdue.edu+5.
- **URL:** https://arxiv.org/abs/2402.01763